Class #1 - A Linear System

1. Recall: Linear function from Calculus $f(x) = 2x$
   Note:

   (a) $f(0) =$ __________

   (b) $f(cx) =$ __________ for any real number c

   (c) $f(x_1 + x_2) =$ __________ for any x_1 and x_2

   A linear function is a good approximation of a __________function locally.

2. Linear Algebra: We again focus on linear functions but from $\mathbf{R^m} \to \mathbf{R^n}$.
   $\mathbf{R^m}$: set of ordered m-tuples of real numbers.

   Example: a person's (age, heart rate, height, weight, student ID number) $\in$ __________.

   We analyze linear functions from $\mathbf{R^m} \to \mathbf{R^n}$, and use these knowledges for more complicated functions later on.

3. What is a *Linear Equation*?
   Def: A Linear Equation is an equation of the form

   $$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

   where $a_1, a_2, \cdots, a_n$ and b are real numbers, and $x_1, x_2, \cdots, x_n$ are variables.
   We call a_i coefficients.

   Examples:
   $x_1 = 2(\sqrt{13} - x_3)$ is a __________equation.
   $x_2 = 2x_1x_3 + 5$ is __________equation.

   Note: Scaling by a nonzero constant or adding a same constant to both sides of the equation does not change the equation.

4. Def: A *system of linear equations* or *Linear System* is a collection of one or more linear equations involving the same variables.

   Example: $\begin{cases} x_1 - 2x_2 & = & -1 \\ -x_1 + 3x_2 & = & 3 \end{cases}$

   This has the same solution set as the system $\begin{cases} x_1 & = & 3 \\ x_2 & = & 2. \end{cases}$

5. Def: Two linear systems are called *equivalent* if they have the same solution set.

6. Example: Are the following systems equivalent?

$$\text{(I)}\begin{cases} 2x_1 & -2x_2 & & = & -1 \\ -x_1 & +3x_2 & +x_3 & = & 3 \\ & 4x_2 & -2x_3 & = & 1 \end{cases} \qquad \text{(II)}\begin{cases} -x_1 & +3x_2 & +x_3 & = & 3 \\ 2x_1 & -2x_2 & & = & -1 \\ & 4x_2 & -2x_3 & = & 1 \end{cases}$$

$$\text{(III)}\begin{cases} x_1 & -3x_2 & -x_3 & = & -3 \\ 2x_1 & -2x_2 & & = & -1 \\ & 4x_2 & -2x_3 & = & 1 \end{cases} \qquad \text{(IV)}\begin{cases} x_1 & -3x_2 & -x_3 & = & -3 \\ & 4x_2 & +2x_3 & = & 5 \\ & 4x_2 & -2x_3 & = & 1 \end{cases}$$

Observations:

(a) The new system is obtained from the previous system by one of these operations:

(i) ________________________: Replace one row by sum of itself and a multiple of another row

(ii) ________________________: Interchange two rows

(iii) ________________________: Multiply all entries in a row by a nonzero constant

Def: Above three operations are called the *elementary row operations.*

(b) An elementary row operation __________the solution set.

(c) An elementary row operation is __________.

(d) For (I)-(IV), __________and __________are common.

For each system, one can associate two matrices (=array of numbers):

Coefficient matrix and *Augmented matrix.*